A little foresight. Better care.

GlucoAhead

A focused diabetes digital twin
Patient history and sensor signals inform the next hour.

One patient, one specific prediction

A doctor sees the recent glucose trend.
GlucoAhead adds a forecast of the next 60 minutes.

The prototype highlights a possible reading above
180 mg/dL, then lets the reviewer check the outcome.

Reference range: 70–180 mg/dL. Individual targets can differ.

Two data sources enter the same model

Synthetic EHR

Age, BMI, HbA1c,
diabetes duration

Sensor history

Glucose, heart rate,
steps and sleep

+

Feature fusion

22 standardized inputs
Past data only

>

Forecast + replay

12 horizons
Residual band
Compare with future

A working model with a small footprint

Multi-output ridge regression
12 predictions at five-minute intervals

80 patients for training
20 patients for validation residuals
30 unseen patients for testing

Patient-disjoint split. Entirely synthetic data. Seed 20261018.

Lower error on the synthetic benchmark

7.53

Combined model MAE

mg/dL, averaged across 5–60 min

9.44

Persistence baseline

mg/dL, future equals current

20.3%

Relative improvement

against persistence baseline

8,160 test windows. 60-minute model MAE: 12.27 mg/dL.

Spike warnings miss many future events

80.8%

Precision

17.2%

Recall

59 true warnings, 14 false warnings, 285 missed positive windows.
Unknown future meals are a major limitation.
The prototype cannot support clinical decisions.

A demonstration the reviewer can inspect

Patient overview
Play readings and edit EHR inputs to update the twin.

Prediction replay
Freeze the forecast, then reveal and score the future.

Model evidence
Inspect the baseline, ablation, misses and exact results.

A focused foundation for further validation

Delivered
Working dashboard, reproducible model and test suite.
Architecture, presentation, model card and demo script.

Next evidence needed
Real anonymized CGM data, external evaluation,
calibration and clinician review.

Before submission: add team details, public GitHub and demo-video link.